WEEK 3: ASSIGNMENT 3

Q1 Let a population have mean $\mu = 50$ and variance $\sigma^2 = 100$. For samples of size $n = 25$, the variance of the sampling distribution of the mean is:
a) 100
b) 20
c) 4
d) 25

Q2 Which statement about the Central Limit Theorem is FALSE?
a) It applies to the sample mean
b) It requires normal population
c) Approximation improves as n increases
d) Sampling distribution tends to normality

Q3 If $P = 0.6$ and $n = 150$, what is the standard deviation of the sampling distribution of $\hat{p}$?
a) 0.04
b) 0.05
c) 0.06
d) 0.08

Q4 A sample of size 36 has mean 80 and population standard deviation 12. The margin of error for a 95% CI is:
a) 2
b) 3.92
c) 4
d) 6

Q5 Which of the following estimators is unbiased?
a) Sample standard deviation
b) Sample variance
c) Sample range
d) Sample median

Q6 The width of a confidence interval for the population mean will decrease when:
a) Confidence level increases
b) Sample size increases
c) Population variance increases
d) Sample mean increases

Q7 A 95% CI for a mean is (46.5, 51.5). Which statement is correct?
(a) The population mean has a 95% probability of lying in the interval
(b) 95% of sample observations lie in the interval
(c) 95% of confidence intervals constructed in repeated sampling contain μ
(d) The confidence level equals the sample mean

Q8 When σ^2 is unknown and sample size is small, the appropriate distribution for CI estimation of μ is:
a) Z distribution
b) Chi-square distribution
c) F distribution
d) t distribution

Q9 A variable is normally distributed with:
Mean (μ) = 50
Standard deviation (σ) = 10
What is the Z-score for a value of 65?
a) 1.0
b) 1.5
c) 2.0
d) 2.5

Q10 A sample of size 16 is drawn from a normal population. The chi-square statistic for variance estimation has degrees of freedom equal to:
a) 15
b) 16
c) 14
d) 30

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